

Does It Still Make Sense? Organizing and Summarizing Resources in Cross-Session Aggregated Search

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Abstract

When undertaking complex search tasks that cannot be completed in a single session, resuming the search task can be challenging. When the search is conducted over multiple platforms, this challenge increases. While aggregated search interfaces support the synthesis of search results from multiple sources, cross-session support in such settings remains underexplored. In this research, we study a set of novel workspace approaches to this problem, within a digital humanities search context. We introduce an interactive cluster refinement (ICR) workspace that automatically organizes saved resources using thematic clustering, preserves their provenance, and provides an interactive method for refining the cluster to match the searcher's mental model. Building on this foundation, we investigate two augmentations to facilitate reacquaintance with previous search activities: interactive passage highlighting (ICR+IPH), which lets searchers mark salient passages as a reminder of why resources were saved; and AI-generated summarization (ICR+AIS), which produces concise overviews of saved resources with inline citations. We combine these together into a fourth interface (ICR+IPH+AIS), where the AI-generated summarization is instructed to make use of the passage highlighting. We conducted a cross-session laboratory study with $n = 36$ participants to compare these interface alternatives with respect to measures of utility, perceived value, metacognition, exploratory search behaviour, and cross-session support. Both ICR+IPH and ICR+AIS improved outcomes relative to ICR. The combined ICR+IPH+AIS interface was most effective in enabling cross-session searching, yielding faster task resumption, higher resource precision, greater perceived value and knowledge gain, and stronger support for metacognitive planning and evaluation as well as exploratory search outcomes.

CCS Concepts

• **Information systems** → **Combination, fusion and federated search; Digital libraries and archives; Search interfaces; Users and interactive retrieval.**

Keywords

Exploratory search, Search task resumption, Reacquaintance, GenAI summarization, Sensemaking, Digital humanities

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1 Introduction

Searching a common topic across multiple platforms is challenging in domains such as the digital humanities, where relevant materials are scattered across archives, libraries, and encyclopedias [11]. Aggregated search addresses this fragmentation by unifying results from heterogeneous sources [6, 7, 9]. For complex information needs, these activities routinely span multiple sessions [32, 79], making task resumption a key requirement. However, the added heterogeneity and volume introduced by aggregation can hinder efficient resumption unless interfaces provide explicit cross-session support [8, 31, 38, 40, 41].

Traditional search interfaces treat searches as isolated activities, lacking structured support for task resumption beyond basic bookmarks, history lists, or tabs. Such basic artifacts record what searchers found, but not why particular resources mattered or how subtopics and decision points emerged, leaving the cognitive work of reconstruction largely to the searcher [56]. When systems lack structured support for cross-session reacquaintance, searchers frequently repeat queries, revisit previously examined resources, and stop making progress, leading to redundancy and frustration [39, 40]. Memory for prior actions and rationales fades, and the task's conceptual structure likely has not been externalized into stable artifacts that can be efficiently resumed [45].

Building on prior work that aggregates search results across multiple digital humanities platforms into a single, provenance-preserving environment [55], we propose an interactive aggregated workspace to support sensemaking amid cross-session searching. This workspace provides a unified environment where searchers can save, organize, and revisit saved resources. It incorporates foundational capabilities for automated topical clustering and interactive drag-and-drop reorganization and refinement of the clusters, allowing searchers to structure their findings intuitively. By preserving the provenance of each saved resource and externalizing the searcher's organizational framework, the workspace aims to support sensemaking of found resources, reduce reorientation costs, and enable smoother task resumption across sessions.

We also consider two complementary task-resumption scaffolds within the workspace: interactive passage highlighting, which lets searchers mark salient passages in saved resources so those cues



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persist as reminders when they return to the task, and AI-generated summarization, which provides concise, provenance-aware cluster overviews that can be read directly in the workspace and at resumption, allowing searchers to scan a cluster rather than re-opening each saved resource one by one.

Accordingly, we conducted a controlled between-subjects laboratory study in a digital humanities setting that compares four alternatives for an aggregated search workspace (composed of combinations of novel task resumption features). Each participant used one assigned interface across two sessions (an initial session and a resumed session separated by one week) allowing us to assess how well the designs support task resumption. The evaluation focuses on behavioural performance, participants' subjective experience, and analysis of interaction logs, providing a comparative view of how the interfaces support reacquaintance and progress across sessions.

2 Literature Review

Complex search tasks with ill-defined, multi-faceted information needs [79] often lead to people undertaking exploratory search strategies spanning multiple sessions [3, 10, 47, 56]. Exploratory search is an iterative process of learning and investigation [32, 79]. Foundational models describe a two-phase process: an initial period of broad, exploratory browsing style of searching for discovery and learning, which transitions to focused searching involving query reformulation and information extraction once the searcher's uncertainty is reduced [79].

The multi-session nature of exploratory search introduces significant resumption challenges. Returning to a suspended task is cognitively taxing, as searchers must reconstruct their prior context manually from memory in the absence of interface support [45]. To cope, searchers adopt reacquaintance strategies, including re-issuing prior queries, reviewing saved artifacts, and skimming earlier results [14, 41]. Insufficient support for the reacquaintance process often results in redundant efforts and searcher frustration [39].

Traditional interfaces that treat searches as isolated activities, such as linear histories or bookmarks, are insufficient for resuming complex tasks [45]. More advanced supports include mechanisms for showing past search activity. SearchBar [56] uses a collapsible tree of topics and queries to help searchers resume their task via process cues. LogCanvas [82] and Search Timelines [31] use timelines to surface the temporal structure of search activities. Workspaces provide support for saving and organizing search results, by providing note-taking capabilities (OrgBox [18]), multi-workspace structures (Dilex Search [25]), or nested workspace structures (WE-Search [59]). While these approaches provide valuable mechanisms to assess past search activities when returning to a search task, they are all focused on searching within a single platform.

The resumption challenge is magnified in fields like the digital humanities, where searchers must navigate a fragmented landscape of disparate archives, academic libraries, and encyclopedias. Aggregated search interfaces that query multiple sources simultaneously can reduce the burden of platform switching [6, 53, 55], providing explicit support for resuming search tasks remains a need [54]. For scholarly work, preserving provenance (the source identity of each

resource) is essential for critical appraisal, establishing trust, and proper citation [15].

Fundamentally, complex search is a sensemaking process of constructing a conceptual framework to bridge a knowledge gap [62]. To manage the cognitive demands of this process, searchers rely on externalizations, which involve offloading their evolving understanding into representations like notes, categories, or diagrams [29, 67]. These external artifacts act as a stable "external memory", supporting iterative foraging and schema formation by making the searcher's thinking visible and manipulable [85]. Comprehensive models of sensemaking confirm that encoding information into such representations and adjusting them based on new insights are essential elements of the process [86–88], a need reinforced by evidence that limited working memory otherwise impedes the integration of information during a search session [17].

Presenting search results in a flat, ranked list imposes a high cognitive load by forcing searchers to mentally group related items and to assess the relevance of each result one-by-one [32]. Search result clustering offers a solution by automatically organizing resources into topically coherent groups [20]. Such visual organization aligns the interface's external representation with the searcher's internal mental model of the topic by leveraging perceptual principles, such as using common regions to delineate groups [61, 77].

Recent advances in large language models (LLMs) and generative AI (GenAI) offer tools to manage cognitive load, particularly through automated, citation-grounded summarization [42]. Concise summaries can provide the gist of a collection of resources, but automation introduces a tension between efficiency and user agency; moreover, abstractive summarization can introduce factual errors and omissions [50]. To be effective, human-AI interaction should prioritize searcher control, transparency, and traceability via explicit citations and provenance [5, 63, 70]. Retrieval-augmented generation can further improve groundedness by linking outputs to source passages [37]; workspace-augmented generation extends this concept to search results explicitly saved by the searcher [59]. An interactive approach, where user-generated cues (highlights, notes, pinned resources) steer GenAI output, can better align results with searcher intent and support efficient resumption while keeping the searcher in control [89, 90].

3 Approach: Search Interface Designs

To investigate how different search interface features might support cross-session aggregated search, we first designed a workspace integrated with an existing aggregated search framework [55]. Search queries are issued in parallel to Europeana, our University's Academic Library, and Wikipedia via their respective APIs [23, 24, 51], returning ten resources per platform per page which are organized in clusters based on the textual contents of their metadata. Each search result card preserves source provenance and provides a *save* button to add it to the workspace.

While modern approaches to workspace design provide the search results in an ordered list based on when they were saved, surfacing the timestamp and source query to support the searcher in reacquainting themselves with the past search activities [25], to this we added source platform logos to preserve the provenance of the saved resources. From this foundation, we considered adding

three other approaches that could support sensemaking of the saved resources: organization, highlighting, and summarization. As pursuing a study of all 24 permutations of these four alternatives would be infeasible, we chose four combinations that we considered most promising. These are explained in the subsections that follow.

3.1 Interactive Cluster Refinement (ICR)

The *ICR* interface (Figure 1a) is designed to help searchers organize saved resources into meaningful structures that are aligned with their mental models. The workspace automatically groups saved resources into thematic clusters ①. This organization is achieved through a bottom-up process: titles and descriptive snippets are encoded with sentence-level embeddings (Sentence-BERT) [22, 65] and grouped using agglomerative hierarchical clustering [43]. To provide succinct, human-readable labels for these clusters ②, representative keywords are extracted using a lightweight KeyBERT procedure [26].

The clustering is dynamic, following an incremental update process. When a searcher saves a new resource, its content is encoded into a semantic vector $\vec{v}_{\text{new}}$. This vector is then compared against the centroid $\vec{c}_i$ (the mean vector) of each existing cluster C_i [46]. The resource is provisionally assigned to the cluster C_k that yields the highest cosine similarity: $k = \arg \max_i \cos(\vec{v}_{\text{new}}, \vec{c}_i)$. If this maximum similarity falls below a predefined threshold θ , the resource is considered thematically novel and seeds a new singleton cluster. Otherwise, it is added to cluster C_k , and the cluster's centroid $\vec{c}_k$ is recalculated to incorporate the new resource [2]. To help searchers assess the recently saved resources in the workspace, clusters are ordered to prioritize recent activity.

We chose an incremental approach to updating the clusters to preserve the user's mental model of the workspace. While global re-clustering may produce better quality clusters, they do so at the expense of introducing semantic flicker [52], characterized by the abrupt splitting or merging of established clusters. By favouring incremental updates over cluster optimization, the workspace keeps previously generated clusters intact, providing reliable and persistent externalization of the user's cognitive map.

While this approach to clustering saved resources provides a useful default, we prioritize searcher agency and alignment with their evolving task frames [71, 84]. To do so, searchers are given full interactive control over the clustered organization. They can organize saved resources within a cluster as well as merge and split clusters by dragging-and-dropping the saved resource card ③. They can re-order entire clusters via drag handles ④, and rename cluster labels with in-place editing ⑤. Preserving the searcher's spatial layout leverages spatial memory and externalized organization, which have long been shown to aid re-finding and sensemaking in document workspaces and spatial hypertext [49, 66].

The key benefit of this approach to workspace organization of saved resources is that it externalizes the searcher's evolving cognitive map (emerging categories, rationales, and cues) so that task resumption after a pause is guided by stable, manipulable representations [67, 88]. By offloading memory and aligning visible structure with the searcher's conceptual organization, such representations can mitigate working-memory limits and support faster, more confident resumption [17].

3.2 Interactive Passage Highlighting (ICR+IPH)

The *ICR+IPH* interface (Figure 1b) supplements the *ICR* interface with the ability to interactively highlight passages of specific interest to the searcher. This is initiated by clicking the Highlight Passages button ⑥ on a resource, which opens a dialogue box allowing searchers to select and highlight text from its title or body text, as well as remove existing highlights. These highlights serve multiple purposes: they act as personal annotations of individual saved resources, and serve as summaries of the entire workspace ⑦ as well as summaries of the resources in specific clusters ⑧. Unlike other work that uses highlighting to reveal relationships across saved resources [32], this approach treats highlighting as a discrete action on the specific saved resource only.

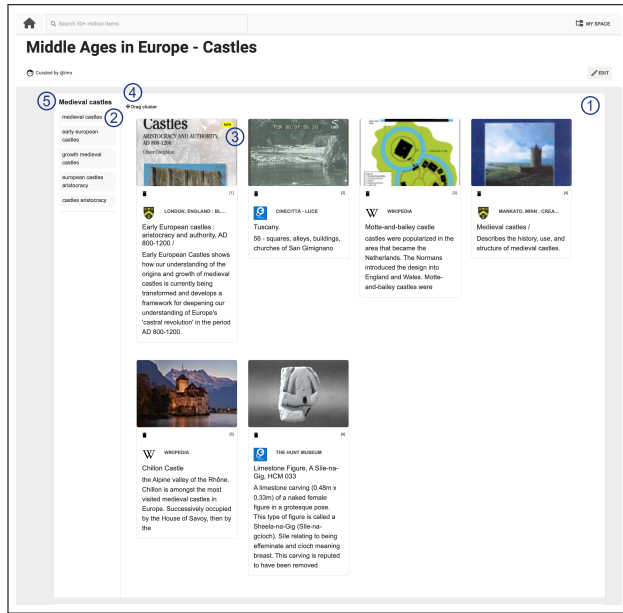
This interactive passage highlighting has been designed to help searchers externalize what they deem salient for specific saved resources, which holds promise for improving comprehension and information reacquisition [48]. It provides a mechanism for personalizing the workspace [21, 71] in a way that can be easily updated as the searcher's mental model changes. For cross-resources sensemaking, preserving and aggregating user-marked snippets may ease comparison and connection-making without full re-reading of the resource [27]. Persisting those marks across sessions may provide personally meaningful cues for why a resource was saved, aligning with evidence collection in sensemaking while maintaining agency [17].

3.3 AI-Generated Summarization (ICR+AIS)

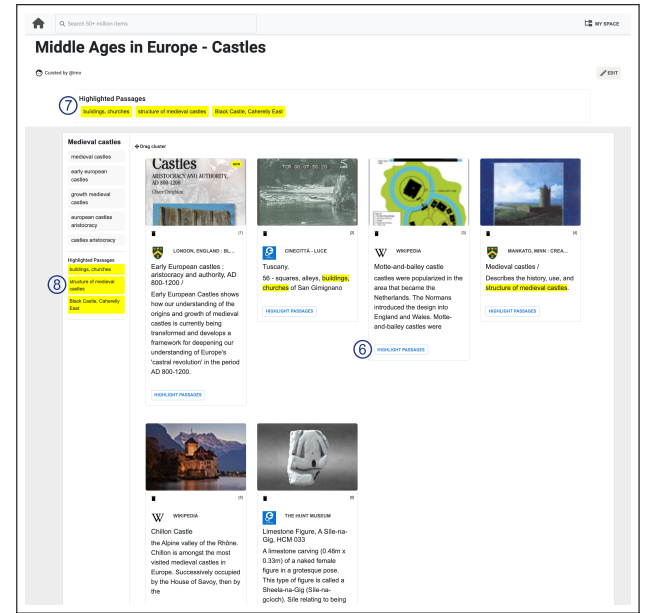
The *ICR+AIS* interface (Figure 1c) augments the *ICR* interface by adding attributable AI-generated summaries. Two levels of GPT-4-generated [1] summaries are provided to support searchers in reacquainting themselves with previously saved resources: workspace overview ⑨ and cluster summaries ⑩. The generation relies on a structured, two-part prompting scheme [68, 78] that assigns the model a “concise, domain-expert summariser” system role to establish a consistent persona [91]. The prompt's user message serializes the current spatial arrangement of resources: clusters are listed top-to-bottom, and resources within each cluster appear in row-first order. The instructions direct the model to prioritize earlier-listed evidence, acknowledging known positional biases in long-context models [16, 44, 64]. For long inputs, a hierarchical pass (summarizing chunks, then generating the summaries) preserves attribution [81], and a low generation *temperature=0.4* ensures stable outputs [33].

This process produces separate bulleted summaries for each cluster and for the overall workspace. They are neutral in tone and include inline citations for traceability [63]; each point in the bulleted list has a bracketed reference to the numbered saved resource. To support quick assessment of the information, we targeted the summaries to be readable within one minute. Given a typical adult silent reading rate of 238 words per minute [13], the summaries were capped at 250 words. Summaries automatically re-generate as searchers save new resources or manipulate cluster compositions.

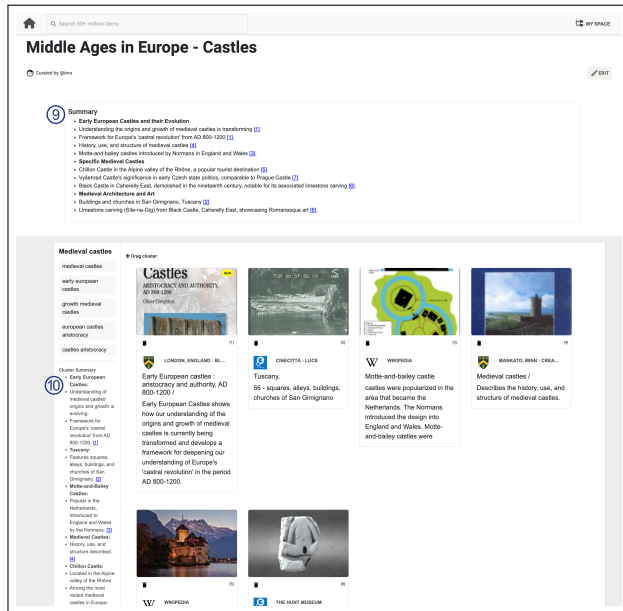
The concise and attributable overviews have been designed to provide the gist needed to refresh situation awareness without exhaustive re-reading, which is expected to support efficient task resumption, especially when the tasks are complex in nature [4, 75].



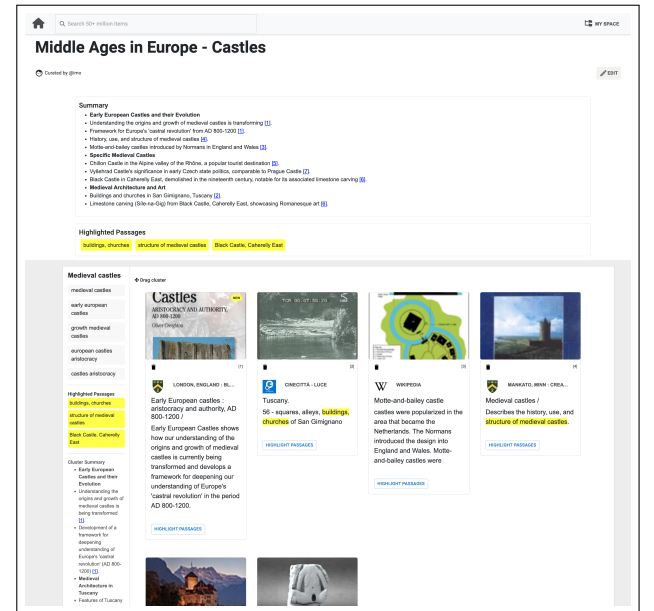
(a) The Interactive Cluster Refinement (ICR) interface.



(b) The Interactive Passage Highlighting (ICR+IPH) interface.



(c) The AI-Generated Summarization (ICR+AIS) interface.



(d) The Combined (ICR+IPH+AIS) interface.

Figure 1: The four experimental interface designs.

Presenting summaries at both workspace and cluster levels supports sensemaking at multiple granularities, while preserving the link to the resources for verification [69]. Limiting the length is intended to minimize the extra cognitive load of having to read yet more text in the search interface, promote skimmability, and serving a function similar to that of scholarly abstracts, which enable rapid evaluation of content [13, 60, 72].

3.4 Combined (ICR+IPH+AIS)

The *ICR+IPH+AIS* interface (Figure 1d) integrates the functionalities of all preceding conditions to support human-AI collaboration. In this design, user-generated highlights ⑥–⑧ function as explicit signals of informational salience. These annotations are systematically incorporated into the generative model’s prompt as an explicit “salience clause” [83]. This clause instructs the GenAI to produce

summaries ⑨–⑩ to prioritize the specific entities, claims, and phrases the searcher has marked as important, while still considering the full context of the unmarked evidence.

Combining explicit human cues with automated synthesis is intended to leverage complementary strengths, where human judgment specifies salience and the model performs compression [46]. Mixed-initiative findings suggest that steering summaries with workspace cues or user-provided signals could improve alignment and perceived usefulness for sensemaking [73]. Embedding highlighted passages into the prompt operationalizes human-in-the-loop principles [5, 28], with the potential to preserve agency and transparency while reducing the effort of reconstituting context when returning to a previous search activity. This synergy yields summaries that are quick to scan and are tailored to the searchers' evolving interpretations of their information spaces.

In order to enable the reproducibility of this research, the source code for the project has been posted to GitHub, including details of the prompt design and other technical aspects of the approaches ¹.

4 Evaluation

This evaluation focuses on cross-session aggregated search, examining how searchers performed over two distinct phases: an initial search session and a subsequent resumed session. The study was guided by the following research questions:

RQ1 (Utility Measures): What are the differences in efficiency, effectiveness, and feature use across the four interface conditions during the initial and resumed sessions?

RQ2 (Perceived Value Measures): How do perceived usefulness, ease of use, satisfaction, engagement, and perceived knowledge differ by interface condition across the two sessions?

RQ3 (Metacognition Measures): How does the interface condition affect planning, monitoring, and evaluation of the search process across the two sessions?

RQ4 (Exploratory Search): To what extent do the interfaces support exploratory search behaviour across the two sessions?

RQ5 (Cross-Session Search): How well do the interfaces support task preparation at the end of the initial session and task resumption at the start of the resumed session?

4.1 Study Design

To assess how different interface conditions support cross-session exploratory search in an academic digital humanities context, a controlled laboratory study was designed using a between-subjects design [34]. The four interface variants described in Section 3 serve as the experimental conditions: **ICR**, **ICR+IPH**, **ICR+AIS**, and **ICR+IPH+AIS**. Each participant completed the two separate search sessions using an assigned interface. This enabled analysis of how the interface condition affected reacquaintance and continuation in cross-session search.

For addressing research question **RQ1 (Utility)**, we measured efficiency (time to task completion for each session, along with time to task reacquaintance in the resumed session, which was operationalized as the interval from when the workspace loaded to the first logged search activity, indicating task resumption) and

effectiveness (precision of saved resources). Frequency of unique feature use was derived from interaction logs.

For **RQ2 (Perceived Value)**, we measured several constructs. For overall measures of usability, we used TAM2-based scales for perceived usefulness and ease of use [34, 76], and a satisfaction scale modelled after TAM2 [32]. We assessed user engagement, an important component of the overall user experience [57], using the User Engagement Scale-Short Form (UES-SF) [58]. This scale is composed of four dimensions: focused attention, perceived usability, aesthetic appeal, and reward factor. These data were collected post-task for each session. In addition, pre- and post-task measures were collected in each session using a four-item perceived knowledge composite (knowledge, interest, certainty, confidence) [31], enabling the calculation of perceived knowledge gain.

RQ3 (Metacognition) was examined with a 15-item instrument administered post-task for each session, yielding planning (3 items), monitoring (3), and evaluation (9) subscales [19].

To address **RQ4 (Exploratory Search)**, we employed a 14-item instrument covering seven dimensions of exploratory search (discovery, learning, investigation, uncertainty reduction, query reformulation, result examination, and information extraction) adapted from prior exploratory search studies [30, 32]. This instrument was administered post-task for each session.

For **RQ5 (Cross-session Search)**, we administered a six-item instrument post-task in the resumed session, capturing two subscales: task preparation (TP) and task resumption (TR), based on prior work on cross-session searching [80]. All survey data were collected using 5-point Likert scales and coded as specified for each scale.

4.2 Simulated Search Tasks

We employed a simulated work-task methodology [12] to create a complex search task that elicits exploratory search behaviour and cross-session searching. In consultation with a historian and a university librarian, we designed a two-session scenario for a third-year undergraduate seminar on European history. The task was engaging, open-ended, and appropriately difficult [35], with sufficient coverage across Europeana, the academic library, and Wikipedia. The scenario, detailed in Figure 2, required participants to gather academic resources for a class assignment. The initial session focused on broad, exploratory browsing and learning, where participants organized their workspace for a later return. The resumed session centred on reacquaintance with the prior search activity, a transition to focused searching, and continuing to use the workspace for reorganization and pruning.

4.3 Procedures

The study consisted of two in-person sessions separated by one week. This gap was designed to create a realistic scenario where a complex search task must be paused and later resumed, with between-session duration consistent with prior work in cross-session search [14, 25, 56].

Upon arrival for the initial session, participants provided informed consent and completed a pre-study questionnaire. To familiarize participants with the system, they received a brief introduction to the SERP and workspace interfaces and completed a

¹<https://github.com/milad1372/Cross-Session-Aggregated-Search-Workspace>

Search Scenario. You are a third-year university student enrolled in a seminar course on “Issues in European History” that requires a research paper. From the professor’s list of potential topics, you choose *the evolution of European maps*. The professor clarifies that this topic concerns changing European boundaries and political landscapes over time, with examples such as the expansion and contraction of the Roman Empire, the fragmentation of medieval kingdoms, and modern developments like the formation of the European Union.

Your professor emphasizes using diverse, trustworthy academic resources rather than generic web pages, and recommends a new search interface that aggregates results from the Europeana digital humanities archive, the University Academic Library (Ex Libris Primo), and Wikipedia. You know that, beyond aggregation, the interface allows you to save resources into workspaces and organize what you have found to support later resumption.

Initial Session Instructions. Repeat the scenario in your own words to confirm understanding. Open the search interface, create a new workspace for this activity, and spend **10 minutes** searching broadly to learn about different aspects of the topic, followed by **5 minutes** organizing the saved resources in the workspace so your progress will make sense when you return later.

Resumed Session Instructions. Repeat the scenario in your own words to confirm understanding. Open the search interface, select last week’s workspace for this activity, briefly review your saved and organized materials to reacquaint yourself, then choose a specific facet of the topic and continue with a more focused search. Use the workspace at any time to reorganize, add new resources, or remove items that are no longer relevant.

Figure 2: Cross-session simulated work task and instructions.

short guided training exercise to practice issuing queries, viewing resource details, saving resources, organizing the workspace, and using the key features of the assigned interface.

Participants began the initial session with 10 minutes of exploratory browsing, followed by 5 minutes dedicated to organizing their findings to facilitate later resumption. Immediately after the task, they completed the post-task questionnaires.

One week later, the resumed session began with a pre-session questionnaire and a brief training refresher that reviewed the key features of the assigned interface. The participants then undertook a reacquaintance period to review their saved materials before continuing with more focused searching and optional reorganization. After finishing the task, participants completed the post-task and post-study questionnaires. All sessions were conducted in the same lab using an identical 32-inch desktop workstation setup with a consistent browser window size.

The entire study was reviewed and approved by our university’s Research Ethics Board. Note that this paper is part of a larger study that included additional data collection using eye-tracking glasses, screen recordings, and a retrospective think-aloud protocol. This aspect of the study is not included in this paper due to space limitations, and will be reported in a subsequent publication.

4.4 Participants

Participants were recruited using a convenience sampling method among undergraduate students at the University of Regina and its

federated colleges who were currently enrolled in or had completed upper-level European history courses. We used a combination of classroom announcements and email invitations sent to relevant departments. In the recruitment materials, students were invited to participate in the study and were informed that they would receive a \$35 coffee shop gift card per session (\$70 total) as compensation for their time.

Participants were assigned to the four experimental conditions via a round-robin procedure ($n = 9$ per condition). The cohort was balanced across year levels, comprising 12 second-year, 15 third-year, and 9 fourth-year students, all of whom had completed 200- or 300-level European history courses. Twenty-two participants identified as female and fourteen as male. The mean age was 23.4 years ($SD = 2.1$). No participants reported colour-vision deficiencies.

4.5 Data Analysis

The analysis consisted of statistical comparisons between the four interface conditions. Initial checks verified pre-task equivalence across groups and found no association between demographic characteristics and outcomes, supporting an analysis focused on the interface factor without covariates. The data was aggregated by concept: numerical measures of utility (RQ1), and Likert-scale composites for perceived value (RQ2), metacognition (RQ3), exploratory behaviours (RQ4), and cross-session support (RQ5).

To compute precision, two independent raters assessed the relevance of all saved documents. They used a 3-point relevance scale (not/partial/full) and achieved high agreement ($\kappa = 0.94$). This data was converted into a binary measure, where partially and fully relevant items were considered ‘relevant’ to align with the exploratory nature of the search activity. The calculation of the precision score followed the standard approach [34].

All composites were formed after internal consistency was confirmed with Cronbach’s α . The data was assessed for normality and homogeneity of variance to select the appropriate statistical test. A p -value of 0.05 was used to determine statistical significance.

For the numerical data and most Likert composites (RQ1, RQ2, RQ4, RQ5), distributions met the assumptions of normality. One-way ANOVAs were used, followed by Tukey’s HSD for post-hoc analysis. Partial eta-squared was used for effect size measurements, interpreted as small (0.01), medium (0.06) and large (0.14) [36].

For the analysis of the metacognition data (RQ3), assumptions of normality and homogeneity of variance were not met. Non-parametric Kruskal-Wallis H tests were used, with Dunn-Bonferroni post-hoc analysis. Epsilon-squared was used to measure effect size, interpreted as small (0.01), medium (0.08) and large (0.26) [74].

In the section that follows, results are reported in tabular format. Bold font signifies that the criteria for statistical significance was met and that the effect size was large. Post-hoc analysis outcomes are marked using a single-asterisk (*) to signify conditions where statistical significance was found versus at least one other factor, and a double-asterisk (**) to signify conditions where statistically significance was found versus all other conditions. When discussing these findings, the p -level for the post-hoc analysis is provided.

5 Results

5.1 RQ1: Utility Measures

5.1.1 Initial Session. During the initial session, task efficiency was comparable across all conditions, with no significant difference in the time to task completion (see Table 1, left). In contrast, significant differences in effectiveness were found: the precision of saved resources was significantly higher for the ICR+IPH ($p < 0.01$) and ICR+AIS ($p < 0.01$) conditions compared to ICR, while the precision for those in the ICR+IPH+AIS condition was significantly higher than all other conditions ($p < 0.01$). Providing searchers with automatic (ICR+AIS) and interactive (ICR+IPH) mechanism for assessing what was found helped ensure the accuracy of the saved search results, with the combination of these (ICR+IPH+AIS) providing the greatest benefit.

These effectiveness gains are reflected in interaction patterns. Participants in the ICR+AIS condition engaged in cluster refinement most frequently ($p < 0.001$), suggesting that the AI-generated summaries helped identify and correct organizational misalignment. Those in the ICR+IPH condition used cluster refinement significantly less than all other conditions ($p < 0.001$), which was made up for in use of the passage highlighting feature. Participants who could both highlight passages and read summaries (ICR+IPH+AIS) highlighted the most passages in the saved resources ($p < 0.01$), illustrating that using these in the AI-generated summaries encouraged active engagement. Those who had access to interactive passage highlighting (ICR+IPH and ICR+IPH+AIS) went deeper into the search results lists through pagination ($p < 0.01$), indicating that actively considering what aspects of saved resources are relevant motivated participants to look at more results.

5.1.2 Resumed Session. In the resumed session, adding support to the interactive cluster refinement interface resulted in significant performance differences (see Table 1, right). The time participants took to acquaint themselves with prior work improved in the ICR+IPH ($p < 0.01$) and ICR+AIS ($p < 0.05$) conditions, with the quickest time in ICR+IPH+AIS ($p < 0.001$), indicating skimming rather than rereading. This efficiency was mirrored in time to task completion, with those who had access to all features finishing the task the quickest ($p < 0.01$). The pattern of effectiveness in saving relevant resources mirrored the initial session: ICR+IPH ($p < 0.001$) and ICR+AIS ($p < 0.001$) exceeded ICR, and ICR+IPH+AIS ($p < 0.001$) exceeded all others.

While some behavioural patterns were the same, the differences are noteworthy. Those with AI-generated summaries (ICR+AIS and ICR+IPH+AIS) saved more resources in the resumed session ($p < 0.01$), indicating summaries supported more effective re-engagement after acquainting with prior activities. This was also reflected in more active engagement with interactive features, with ICR+IPH+AIS the most active in refining clusters ($p < 0.001$) and highlighting passages ($p < 0.01$).

5.2 RQ2: Perceived Value Measures

5.2.1 Initial Session. Patterns were found among aspects of perceived value across conditions (see Table 2, left). There was no difference between ICR and ICR+IPH, which shows that interactive passage highlighting on its own was not adding much value during the

exploratory browsing aspect of the initial session. Those in the condition that included AI-generated summaries (ICR+AIS) reported higher perceived usefulness ($p < 0.05$), satisfaction ($p < 0.001$), focused attention ($p < 0.01$), and reward factor ($p < 0.05$), but at the expense of reduced ease of use ($p < 0.05$). That is, the summaries added value, but the extra information added to the interface made it more difficult to use. Those who had access to all three features (ICR+IPH+AIS) reported higher responses than all other conditions for perceived usefulness ($p < 0.001$), satisfaction ($p < 0.001$), focused attention ($p < 0.01$), and reward factor ($p < 0.05$).

Overall, this shows that interactive passage highlighting became valuable when it enhanced the AI-generated summaries, further increasing the value of the summaries. The value of the AI-generated summaries was also reflected in the change in perceived knowledge (see Table 3, left), with significant increases for those who used ICR+AIS and ICR+IPH+AIS. Comparing perceived knowledge gain between experimental conditions also revealed significant differences between these two conditions and the two that did not include the AI-generated summaries ($p < 0.05$).

5.2.2 Resumed Session. The perceived value showed a different pattern after the resumed session for ICR+IPH, and a similar pattern for ICR+AIS and ICR+IPH+AIS (see Table 2, right). Those who had access to interactive passage highlighting reported higher perceived usefulness ($p < 0.001$), satisfaction ($p < 0.001$), focused attention ($p < 0.001$), perceived usability ($p < 0.001$), and reward factor ($p < 0.001$) compared to the ICR condition. This was not the case for the initial session; what changed was the reacquaintance with prior search activities and the instructions to focus the search on a specific facet. Results were similar for ICR+AIS, with the exception that AI summaries continued to have the lowest ease of use scores ($p < 0.05$). For those who had access to all the features (ICR+IPH+AIS), scores on most measures were higher than all other conditions: perceived usefulness ($p < 0.001$), satisfaction ($p < 0.001$), focused attention ($p < 0.001$), perceived usability ($p < 0.001$), and reward factor ($p < 0.001$).

This highlights the value the combination of approaches provided to the overall usability and user engagement. The change in perceived knowledge mirrored these findings (see Table 3, right). Comparing perceived knowledge gain between experimental conditions, those who used ICR+IPH and ICR+AIS reported greater knowledge gains than ICR ($p < 0.01$ and $p < 0.001$, respectively), and those who used ICR+IPH+AIS reported greater knowledge gains than ICR ($p < 0.001$) and ICR+IPH ($p < 0.05$). The cumulative effect of interactive passage highlighting influencing the AI-generated summaries had the greatest impact on the perception of knowledge gain among the participants.

5.3 RQ3: Metacognition Measures

5.3.1 Initial Session. Significant differences were observed across all metacognitive measures during the initial session (see Table 4, left). For planning, participants who had access to AI-generated summaries (ICR+AIS) reported higher scores ($p < 0.01$); those who also had access to interactive passage highlighting (ICR+IPH+AIS) reported the highest scores ($p < 0.01$). A similar pattern was found for monitoring, where conditions that included AI-generated summaries (ICR+AIS and ICR+IPH+AIS) led to significantly higher

Table 1: Mean values (standard deviation) for the utility measures collected during both the *initial* and *resumed* sessions.

Measure / Feature	Initial Session					Resumed Session				
	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA
Time to task reacquaintance	–	–	–	–	–	1:30 (0:40)	1:10 (0:35)*	0:55 (0:30)*	0:40 (0:25)**	$F(3, 32) = 9.21, p < 0.001, \eta_p^2 = 0.46$
Time to task completion	14:58 (0:32)	14:50 (0:35)	14:45 (0:30)	14:40 (0:28)	$F(3, 32) = 0.81, p = 0.49$	17:30 (0:20)	16:40 (0:40)	16:05 (0:45)*	15:45 (0:50)**	$F(3, 32) = 5.13, p < 0.01, \eta_p^2 = 0.32$
Precision of saved resources	0.72 (0.12)	0.77 (0.11)*	0.81 (0.10)*	0.85 (0.08)**	$F(3, 32) = 6.92, p < 0.01, \eta_p^2 = 0.39$	0.74 (0.11)	0.80 (0.10)*	0.84 (0.09)*	0.88 (0.07)**	$F(3, 32) = 8.65, p < 0.001, \eta_p^2 = 0.45$
Issue query	4.0 (1.3)	4.1 (1.2)	4.2 (1.1)	4.0 (1.0)	$F(3, 32) = 0.22, p = 0.88$	3.5 (1.2)	3.8 (1.1)	4.0 (1.0)	3.9 (1.2)	$F(3, 32) = 0.95, p = 0.43$
View resource	6.5 (1.5)	7.0 (1.4)	7.2 (1.3)*	7.3 (1.2)*	$F(3, 32) = 4.51, p < 0.01, \eta_p^2 = 0.30$	6.0 (1.6)	7.5 (1.5)*	7.6 (1.3)*	7.8 (1.2)**	$F(3, 32) = 6.23, p < 0.01, \eta_p^2 = 0.37$
Use pagination	2.0 (0.8)	3.2 (0.9)*	2.1 (0.8)	3.5 (1.0)*	$F(3, 32) = 7.12, p < 0.01, \eta_p^2 = 0.40$	1.8 (0.7)	3.0 (1.0)*	2.0 (0.8)	3.6 (1.1)**	$F(3, 32) = 8.12, p < 0.001, \eta_p^2 = 0.43$
Save resource	5.0 (1.5)	5.0 (1.4)	5.2 (1.6)	5.5 (1.8)	$F(3, 32) = 0.89, p = 0.45$	4.5 (1.4)	4.5 (1.5)	6.0 (1.6)*	6.0 (1.7)*	$F(3, 32) = 4.78, p < 0.01, \eta_p^2 = 0.31$
Delete resource	1.0 (0.5)	1.0 (0.6)	1.1 (0.5)	1.0 (0.7)	$F(3, 32) = 0.31, p = 0.82$	0.9 (0.4)	1.0 (0.5)	1.2 (0.6)	1.0 (0.5)	$F(3, 32) = 0.76, p = 0.52$
Switch SERP/workspace	2.0 (0.9)	2.0 (1.0)	2.2 (1.1)	2.0 (0.9)	$F(3, 32) = 0.45, p = 0.72$	1.8 (0.8)	2.0 (0.9)	2.3 (1.0)	2.0 (0.9)	$F(3, 32) = 0.91, p = 0.44$
Refine cluster	4.0 (1.2)	2.5 (1.1)**	6.5 (1.5)**	4.5 (1.3)	$F(3, 32) = 12.81, p < 0.001, \eta_p^2 = 0.55$	3.8 (1.1)	2.5 (1.0)	7.8 (1.6)*	8.5 (1.7)**	$F(3, 32) = 16.23, p < 0.001, \eta_p^2 = 0.60$
Highlight passage	–	4.0 (1.2)	–	5.8 (1.3)**	$F(1, 16) = 10.12, p < 0.01, \eta_p^2 = 0.39$	–	6.0 (1.2)	–	7.8 (1.3)**	$F(1, 16) = 11.46, p < 0.01, \eta_p^2 = 0.42$

Table 2: Mean values (standard deviation) for perceived usefulness, ease of use, satisfaction, and four aspects of user engagement collected during the *initial* and *resumed* sessions.

Measure	Initial Session					Resumed Session				
	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA
Perceived Usefulness	3.50 (0.65)	3.70 (0.61)	3.90 (0.64)*	4.20 (0.55)**	$F(3, 32) = 8.45, p < 0.001, \eta_p^2 = 0.44$	3.60 (0.66)	3.90 (0.60)*	4.10 (0.57)*	4.40 (0.50)**	$F(3, 32) = 10.12, p < 0.001, \eta_p^2 = 0.48$
Ease of Use	4.20 (0.56)	4.10 (0.58)	3.90 (0.62)*	4.05 (0.60)	$F(3, 32) = 4.12, p < 0.05, \eta_p^2 = 0.28$	4.30 (0.55)	4.20 (0.56)	4.00 (0.60)*	4.18 (0.52)	$F(3, 32) = 3.21, p < 0.05, \eta_p^2 = 0.23$
Satisfaction	3.60 (0.73)	3.80 (0.70)	4.00 (0.65)*	4.30 (0.54)**	$F(3, 32) = 9.20, p < 0.001, \eta_p^2 = 0.46$	3.70 (0.74)	4.00 (0.68)*	4.20 (0.60)*	4.50 (0.48)**	$F(3, 32) = 11.05, p < 0.001, \eta_p^2 = 0.51$
Focused Attention	3.30 (0.82)	3.55 (0.77)	3.75 (0.72)*	4.10 (0.65)**	$F(3, 32) = 5.83, p < 0.01, \eta_p^2 = 0.35$	3.40 (0.80)	3.75 (0.73)*	3.95 (0.68)*	4.30 (0.60)**	$F(3, 32) = 9.12, p < 0.001, \eta_p^2 = 0.46$
Perceived Usability	3.50 (0.78)	3.65 (0.75)	3.80 (0.70)	3.95 (0.68)*	$F(3, 32) = 3.14, p < 0.05, \eta_p^2 = 0.23$	3.60 (0.75)	3.85 (0.71)*	4.05 (0.65)*	4.25 (0.62)**	$F(3, 32) = 7.45, p < 0.001, \eta_p^2 = 0.41$
Aesthetic Appeal	3.80 (0.65)	3.85 (0.63)	3.90 (0.60)	3.95 (0.58)	$F(3, 32) = 0.24, p = 0.87$	3.82 (0.63)	3.88 (0.61)	3.93 (0.59)	3.98 (0.57)	$F(3, 32) = 0.31, p = 0.82$
Reward Factor	3.45 (0.85)	3.65 (0.80)	3.80 (0.75)*	4.05 (0.68)**	$F(3, 32) = 4.21, p < 0.05, \eta_p^2 = 0.28$	3.55 (0.83)	3.85 (0.78)*	4.00 (0.72)*	4.25 (0.65)**	$F(3, 32) = 8.53, p < 0.001, \eta_p^2 = 0.44$

Table 3: Mean values (standard deviation) for perceived knowledge collected before and after the *initial* and *resumed* sessions. Notation: [†] $p < 0.05$, [‡] $p < 0.01$, [§] $p < 0.001$ indicate a significant between-interface pre–post change on Δ .

Measure	Interface	Initial Session				Resumed Session			
		Prior	Post	Δ	Time effect (within)	Prior	Post	Δ	Time effect (within)
Perceived Knowledge	ICR	3.48 (0.75)	3.47 (0.78)	−0.01	$F(1, 8) = 0.14, p = 0.71$	3.57 (0.73)	3.77 (0.72)	0.20	$F(1, 8) = 1.25, p = 0.28$
	ICR+IPH	3.53 (0.71)	3.63 (0.69)	0.10	$F(1, 8) = 2.60, p = 0.14$	3.63 (0.70)	4.18 (0.62)	0.55 [‡]	$F(1, 8) = 10.42, p < 0.01, \eta_p^2 = 0.57$
	ICR+AIS	3.48 (0.76)	3.83 (0.62)	0.35 [†]	$F(1, 8) = 5.83, p < 0.05, \eta_p^2 = 0.42$	3.68 (0.72)	4.42 (0.55)	0.74 [§]	$F(1, 8) = 16.74, p < 0.001, \eta_p^2 = 0.68$
	ICR+IPH+AIS	3.52 (0.74)	3.95 (0.58)	0.43 [†]	$F(1, 8) = 7.42, p < 0.05, \eta_p^2 = 0.48$	3.73 (0.70)	4.54 (0.48)	0.81 [§]	$F(1, 8) = 20.65, p < 0.001, \eta_p^2 = 0.72$

ratings ($p < 0.05$). For evaluation, while adding either interactive passage highlighting (ICR+IPH) or AI-generated summaries (ICR+AIS) resulted in an increase, those who had access to all features (ICR+IPH+AIS) reported the highest level of support for evaluation ($p < 0.05$). These results indicate that additions beyond interactive cluster refinement, particularly the combination of interactive passage highlighting and AI-generated summaries, helped participants plan, monitor, and evaluate during the initial session.

5.3.2 Resumed Session. The benefits of interactive passage highlighting and AI-generated summaries for metacognitive processes were even more pronounced during the resumed session (see Table 4, right). Unlike the initial session, ICR+IPH resulted in statistically significant improvements compared to ICR across all three measures: planning ($p < 0.001$), monitoring ($p < 0.01$), and evaluation ($p < 0.001$). This suggests that interactive passage highlighting was particularly useful after participants reacquainted themselves with previously saved information. ICR+AIS also led to significantly higher scores on all measures. The combination of all features in ICR+IPH+AIS consistently yielded the best results across planning ($p < 0.01$), monitoring ($p < 0.01$), and evaluation ($p < 0.001$). Overall, these findings illustrate that after returning to the search after a one week break, participants appreciated the support provided to reorient themselves, plan next steps, and monitor ongoing progress.

5.4 RQ4: Exploratory Search

5.4.1 Initial Session. While the independent additions of interactive passage highlighting (ICR+IPH) and AI-generated summaries (ICR+AIS) enhanced support for exploratory search activities during the initial session (see Table 5, left), the most significant impact was from the combined approach (ICR+IPH+AIS). For all aspects of exploratory search except query reformulation, participants in ICR+IPH+AIS reported the highest scores with respect to all other conditions: discovery ($p < 0.01$), learning ($p < 0.001$), investigation ($p < 0.01$), uncertainty reduction ($p < 0.001$), result examination ($p < 0.001$), and information extraction ($p < 0.001$). While there was no explicit support for query reformulation, AI-generated summaries influenced by interactive passage highlighting provided an opportunity to identify potential query terms.

5.4.2 Resumed Session. The positive effects on aspects of exploratory search became stronger and more consistent in the resumed session (see Table 5, right). The improvements were amplified across all measures, with support for query reformulation increasing to a level consistent with the other measures. Those in ICR+IPH+AIS reported the highest level of exploratory search support compared to all other conditions: discovery ($p < 0.01$), learning ($p < 0.001$), investigation ($p < 0.01$), uncertainty reduction ($p < 0.001$), query reformulation ($p < 0.01$), result examination ($p < 0.001$), and information extraction ($p < 0.001$). This illustrates

Table 4: Mean values (standard deviation) for the metacognitive measures collected during the *initial* and *resumed* sessions.

Measure	Initial Session					Resumed Session				
	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	H, p, ϵ_H^2	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	H, p, ϵ_H^2
Planning	3.40 (0.70)	3.60 (0.60)	3.90 (0.50)*	4.10 (0.40)**	$H(3)=11.2, p < 0.01, \epsilon_H^2=0.26$	3.50 (0.70)	3.90 (0.60)*	4.20 (0.50)*	4.40 (0.40)**	$H(3)=15.0, p < 0.001, \epsilon_H^2=0.37$
Monitoring	3.30 (0.80)	3.70 (0.70)	3.80 (0.60)*	4.00 (0.50)*	$H(3)=10.5, p < 0.05, \epsilon_H^2=0.29$	3.40 (0.80)	3.90 (0.70)*	4.10 (0.60)*	4.30 (0.50)**	$H(3)=13.2, p < 0.01, \epsilon_H^2=0.31$
Evaluation	3.20 (0.90)	3.60 (0.80)*	3.70 (0.70)*	3.90 (0.60)**	$H(3)=9.6, p < 0.01, \epsilon_H^2=0.27$	3.30 (0.90)	3.80 (0.80)*	4.10 (0.60)*	4.40 (0.50)**	$H(3)=16.8, p < 0.001, \epsilon_H^2=0.43$

Table 5: Mean values (standard deviation) for the exploratory search measures collected during the *initial* and *resumed* sessions.

Measure	Initial Session					Resumed Session				
	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA
Discovery (D)	3.40 (0.70)	3.70 (0.60)*	3.90 (0.50)*	4.20 (0.50)**	$F(3, 32) = 9.10, p < 0.001, \eta_p^2=0.46$	3.50 (0.70)	3.90 (0.60)*	4.20 (0.50)*	4.40 (0.40)**	$F(3, 32) = 12.80, p < 0.001, \eta_p^2=0.55$
Learning (L)	3.30 (0.72)	3.60 (0.62)*	3.90 (0.55)*	4.10 (0.48)**	$F(3, 32) = 10.40, p < 0.001, \eta_p^2=0.49$	3.40 (0.70)	3.90 (0.60)*	4.30 (0.50)*	4.50 (0.40)**	$F(3, 32) = 14.60, p < 0.001, \eta_p^2=0.58$
Investigation (I)	3.50 (0.68)	3.70 (0.60)	3.90 (0.56)*	4.10 (0.50)**	$F(3, 32) = 6.20, p < 0.01, \eta_p^2=0.37$	3.60 (0.70)	4.00 (0.58)*	4.20 (0.52)*	4.40 (0.45)**	$F(3, 32) = 9.90, p < 0.001, \eta_p^2=0.52$
Uncertainty Reduction (UR)	3.20 (0.75)	3.50 (0.65)	3.80 (0.55)*	4.00 (0.50)**	$F(3, 32) = 8.10, p < 0.001, \eta_p^2=0.43$	3.30 (0.72)	3.80 (0.60)*	4.10 (0.50)*	4.30 (0.45)**	$F(3, 32) = 11.70, p < 0.001, \eta_p^2=0.52$
Query Reformulation (QR)	3.60 (0.66)	3.80 (0.60)	3.90 (0.58)	4.10 (0.52)*	$F(3, 32) = 4.20, p < 0.05, \eta_p^2=0.28$	3.50 (0.68)	3.90 (0.60)*	4.00 (0.56)*	4.30 (0.48)**	$F(3, 32) = 7.80, p < 0.001, \eta_p^2=0.42$
Result Examination (RE)	3.70 (0.70)	4.00 (0.60)*	4.10 (0.55)*	4.30 (0.50)**	$F(3, 32) = 7.50, p < 0.001, \eta_p^2=0.41$	3.60 (0.72)	4.10 (0.60)*	4.20 (0.52)*	4.50 (0.45)**	$F(3, 32) = 10.60, p < 0.001, \eta_p^2=0.50$
Information Extraction (IE)	3.40 (0.74)	3.80 (0.62)*	4.00 (0.56)*	4.30 (0.48)**	$F(3, 32) = 9.90, p < 0.001, \eta_p^2=0.48$	3.50 (0.72)	4.00 (0.58)*	4.20 (0.52)*	4.50 (0.45)**	$F(3, 32) = 13.20, p < 0.001, \eta_p^2=0.55$

Table 6: Mean values (standard deviation) for the cross-session search support measures collected after the *resumed* session.

Measure	ICR	ICR+IPH	ICR+AIS	ICR+IPH+AIS	ANOVA
Task Preparation	3.50 (0.65)	3.80 (0.58)*	4.10 (0.52)*	4.30 (0.46)**	$F(3, 32) = 11.21, p < 0.001, \eta_p^2=0.51$
Task Resumption	3.45 (0.70)	3.90 (0.61)*	4.20 (0.50)*	4.50 (0.40)**	$F(3, 32) = 13.60, p < 0.001, \eta_p^2=0.56$

that support enabling task resumption in cross-session searching can also substantially benefit exploratory search, especially for resumed sessions.

5.5 RQ5: Cross-Session Search

At the end of the resumed session, participants reported on support for cross-session searching (see Table 6). These results followed a similar pattern to other subjective measures collected after the resumed session. The highest level of support for task preparation ($p < 0.01$) and task resumption ($p < 0.01$) was reported by those in the ICR+IPH+AIS condition. These results indicate that while each addition to the ICR condition on its own is beneficial for helping searchers reorient themselves, the combination of interactive passage highlighting and AI-generated summaries provided the highest level of support for cross-session search.

6 Discussion

The results provide a consistent pattern of how interactive passage highlighting and AI-generated summaries synergistically support cross-session exploratory search; highlighting gave users direct control to steer the AI summaries. This work extends the literature on controllable summarization, which aims to align summaries with user intent [90]. Whereas prior work controls summaries via explicit attributes like length or topic, we introduce an interactive mechanism where user highlights implicitly steer the AI. This provides a more dynamic and contextual form of control integrated directly into the search workflow.

The combination of these features in the ICR+IPH+AIS interface supported the searchers in collecting highly relevant search results, with efficiency gains realized in the resumed session. Feature use reinforced these roles; by using highlights to curate a more relevant set of passages, searchers reduced the need for manual reorganization and enabled the AI to generate more refined

and targeted summaries. Participants highlighted more in the combined interface, suggesting they found highlighting valuable when it visibly influenced what the summaries reported. These findings align with research arguing that successful AI systems must provide users with agency over automation [28]. Perceived value and metacognitive support also strengthened with the combined features, particularly in the resumed session, as personal cues and condensed overviews lowered reconstruction effort and supported forward planning. This parallels satisfaction gains from steerable summaries in other work [89]. Metacognitive gains are consistent with what has been reported in single-session search activities that provide interface support for interactive note-taking [19].

This work advances the literature on cross-session exploratory search by applying advanced task reacquaintance features to a multi-workspace aggregated search framework. Prior studies have explored workspaces [25, 41] and persistent representations like timelines [31] to help searchers resume tasks, but these have largely been confined to single platform contexts and function as externalized memory that leave the cognitive work of task reconstruction to the searcher [56, 67]. This work provides evidence that when undertaking cross-session exploratory search, providing an external artifact only goes so far to reduce reorientation costs. Instead, we provide evidence of the value in enabling the searcher to actively guide an AI in synthesizing a personalized summary as integrated elements within the external artifact. This moves beyond storing search history to actively supporting sensemaking to enable task resumption.

Furthermore, these findings make an important contribution to the literature on aggregated search, which has primarily focused on the challenges of vertical selection, presentation, and coherence amid increased data volume and heterogeneity [6, 7, 9]. We extend this literature by focusing on how the workspace can be used to support cross-session aggregated search. In particular, we found that user-steered AI summaries are a powerful tool for navigating

the complexity of aggregated results across sessions. By allowing searchers to highlight what is salient, they create a coherent narrative across diverse sources from multiple platforms, directly addressing the coherence challenge.

7 Conclusion

This paper makes three contributions: (1) a problem framing of cross-session exploratory search in the context of aggregating search results across multiple independent search platforms; (2) multiple approaches to provenance-preserving aggregated-search workspaces for the digital humanities, which enable searchers to save resources into clusters, interactively reorganize them via drag-and-drop, highlight key passages within saved resources, and view AI-generated summaries that are anchored to the saved results; and (3) evidence from a laboratory study of cross-session exploratory search, showing how the combination of all proposed features was the superior approach, especially in supporting the activities in the resumed session.

The study was conducted with a convenience sample of undergraduate students, a single simulated work task, and a fixed one-week gap between sessions; ecological validity and generalizability beyond the digital humanities domain remains to be established. Summaries were capped in length and generated with a single LLM configuration; quality is sensitive to underlying metadata and model variability. Similarly, the study's measures focused on short-term resumption, while longer-term knowledge retention and downstream analysis tasks, such as evidence extraction, were outside the scope.

Immediate future work will focus on the analysis of the eye-tracking and retrospective think-aloud data collected in this study, which will provide rich process-level insight into how the interface features shape searchers' cognitive and attentional strategies during cross-session exploratory search. Investigation of other methods for searchers to steer AI-generated content, such as using cluster labels or brief annotations as inputs to the generative model, represents a promising avenue for advancing human-AI collaboration in exploratory search. Further, giving the user more control over the format and structure of the summaries [59] may further enhance the sense of agency. As academic work is increasingly being conducted using mobile devices [25], further study of how these methods for scaffolding cross-session exploratory search processes can benefit mobile searching are needed.

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